

RaktSeva

Blood Bank & Donor Management System

Software Project Management (SPM) | 16-Week Academic Schedule | Parul University

23	20	86.9%	3	16
Total Tasks	Completed	Progress	Team Members	Weeks Total

Overall Project Progress: 86.9%



Project Team

#1 Shuvo Das Lead Architect & Full-Stack Engineer Score: 150 pts Tasks: 9 done	#2 Monami Sadhu Backend & Database Architect Score: 100 pts Tasks: 7 done	#3 Setu Mondol UI/UX & Systems Lead Score: 74 pts Tasks: 4 done
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16-Week Gantt Chart - Project Schedule

RaktSeva Blood Bank & Donor Management System | Software Project Management (SPM)

ID	Task Name	Phase	Assignee	Wk	Prog%	Status	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
PLANNING PHASE																						
GT-01	Project Setup & Scope Definition	Planning	Shuvo Das	2	100%	Done																
GT-02	Hospital Requirement Elicitation & SRS	Planning	Monami Sadhu	2	100%	Done																
GT-03	Feasibility Study & Technology Stack Benchmarking	Planning	Setu Mondol	2	100%	Done																
GT-04	System Scope Finalization & Guide Approval	Planning	Shuvo Das	2	100%	Done																
DESIGN PHASE																						
GT-05	System Architecture, UML & DFD (Level 0, 1, 2)	Design	Monami Sadhu	3	100%	Done																
GT-06	Database Architecture (7 Relational Tables & ERD)	Design	Shuvo Das	2	100%	Done																
GT-07	UI/UX Prototyping (Public Portal & Admin Panel)	Design	Setu Mondol	3	100%	Done																
GT-08	Design Review & Faculty Milestone Sign-off	Design	Monami Sadhu	2	100%	Done																
DEVELOPMENT PHASE																						
GT-09	Core FastAPI Backend Architecture & API Gateway	Development	Shuvo Das	5	100%	Done																
GT-10	User Authentication & bcrypt Security Module	Development	Monami Sadhu	2	100%	Done																
GT-11	Donor Registration & Validation Engine	Development	Shuvo Das	2	100%	Done																
GT-12	Blood Group Search & Compatibility Algorithm	Development	Setu Mondol	2	100%	Done																
GT-13	Executive Admin Panel & Stock Management	Development	Shuvo Das	3	100%	Done																
GT-14	Emergency Blood Request & Query System	Development	Monami Sadhu	2	100%	Done																
GT-15	CMS Page Content Editor & Hospital Master Config	Development	Setu Mondol	2	100%	Done																
GT-16	User Dashboard & Donation History Tracker	Development	Shuvo Das	3	100%	Done																
TESTING PHASE																						
GT-17	Integration Testing & SQLi Security Audit	Testing	Shuvo Das	3	100%	Done																
GT-18	Bug Fixing & Prepared Statements Hardening	Testing	Monami Sadhu	3	100%	Done																
GT-19	Cross-Browser & Mobile UI Responsiveness Testing	Testing	Setu Mondol	3	100%	Done																
GT-20	SPM Thesis Documentation & User Manual	Testing	Setu Mondol	3	98%	Active																
DEPLOYMENT PHASE																						
GT-21	Hospital Staging Deployment & UAT Sign-off	Deployment	Monami Sadhu	2	0%	Pending																
GT-22	Production Maintenance & Database Monitoring Setup	Deployment	Shuvo Das	2	0%	Pending																
GT-23	Academic Project Defense & Final Handover	Deployment	Shuvo Das	2	0%	Pending																
MILESTONES																						
✓	W2: RaktSeva Scope Finalized & SRS Document Approved by Guide	Completed																				
✓	W5: System Architecture, UML Diagrams & 7-Table ER Approved	Completed																				

ID	Task Name	Phase	Assignee	Wk	Prog%	Status	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
V	W9: Core Donor Registration & Blood Search Modules Completed	Completed																				
V	W12: System Testing, OWASP Security Audit & Bug Fixing Cleared	Completed																				
V	W14: Comprehensive SPM Documentation & User Manual Completed	Completed																				
O	W15: Parul Sevashram Hospital Staging Server Setup & UAT Sign-off	In Progress																				
O	W16: Final University Academic Project Defense & Faculty Evaluati...	Upcoming																				

System Modules - RaktSeva Blood Bank

ID	Module Name	Description	Lead	Complete%	Status
1	User Authentication & Role-Based Access	Session authentication, bcrypt 12-round password hashing, JWT bearer tokens, role permissions in users table	Monami Sadhu	100%	Completed
2	Donor Registration & Intake Verification Engine	Public and admin-added donor intake into donor_details table with medical validation (age 18-65, weight >= 50kg, 90-day cooldown)	Shuvo Das	100%	Completed
3	Real-Time Blood Search & Matching Engine	Rapid search across 8 blood groups (A+, B+, AB+, O+, and negative variants) with compatibility matching and max 6 random verified donors	Setu Mondol	100%	Completed
4	User Dashboard & Donation History Tracker	Donor portal, previous donation timeline, next donation eligibility date calculator, and digital donation certificate download	Setu Mondol	100%	Completed
5	Hospital Emergency Blood Request & Query System	Emergency public query tickets handling with Pending/Read/Fulfilled workflow, automated email routing to on-call hospital staff	Monami Sadhu	100%	Completed
6	Executive Admin Management & Audit Portal	Executive Donor CRUD, blood stock inventory oversight, query resolution, hospital contact update in admin_info table	Shuvo Das	100%	Completed
7	Content Management System (CMS) & Public Portal	Rich text editing for public pages: Home, Why Donate, Need Blood, About Us, and Blood Drive Camp scheduler in pages table	Setu Mondol	95%	Completed
8	Hospital Master Configuration & Multi-Branch Directory	Parul Sevashram Hospital master contact info, emergency 24/7 hotline routing, blood bank room numbers, and department directory	Shuvo Das	95%	Completed
9	7-Table Relational Schema & MySQL Optimization	3NF normalized schema: donor_details, users, admin_info, blood, pages, contact_info, contact_query with B-Tree indexes	Monami Sadhu	100%	Completed
10	Security Hardening, Penetration Testing & SPM Defense	Prepared statements against SQL injection, XSS input sanitization, rate-limiting, and university defense documentation	Shuvo Das	95%	Completed

Member Contribution Leaderboard

Rank	Member	Role	Score (pts)	Hours	Tasks Done	Log Count	Share %
1st	Shuvo Das	Lead Architect & Full-Stack Engineer	150	82	9	14	46.3%
2nd	Monami Sadhu	Backend & Database Architect	100	61	7	10	30.9%
3rd	Setu Mondol	UI/UX & Systems Lead	74	45	4	8	22.8%